AIM2 Preparation

Week 2 - Day 2: Pandas Fundamentals

5-Minute Review Notes **|** Series, DataFrames, Selection & Filtering

Day 2 Goal

Learn how Pandas represents tabular data, inspect DataFrames, select rows/columns, filter records, create calculated columns, and generate basic statistical summaries.

Core Idea

Python data -> Pandas DataFrame -> Inspect -> Select -> Filter -> Calculate -> Summarize

1. Import Pandas

```
import pandas as pd
```

pd is the conventional alias used for the Pandas library.

2. Series vs DataFrame

Object	Meaning	Dimensions
Series	One labeled column / 1D data	1D
DataFrame	Table with rows and columns	2D

```
scores = pd.Series([85, 90, 78], index=["Krishna", "Rahul", "Priya"])
```

3. Creating a DataFrame

```
data = {
  "Name": ["Krishna", "Rahul", "Priya"],
  "Math": [85, 75, 91],
  "Python": [92, 80, 95]
}

df = pd.DataFrame(data)
```

A Python dictionary maps naturally to a DataFrame: dictionary keys become column names and lists become column values.

4. Inspecting a DataFrame

```
df.shape
df.columns
df.index
df.dtypes
df.head()
df.tail()
df.info()
df.describe()
```

- **shape** -> number of rows and columns: (rows, columns).
- **columns** -> column labels.
- **index** -> row labels/index.
- **dtypes** -> data type for each column.
- **head()/tail()** -> first/last rows.

- **info()** -> structure, non-null counts, and dtypes.
- **describe()** -> statistical summary of numerical columns.

5. Selecting Columns

```
df["Math"]
df[["Math"]]
df[["Name", "Age"]]
```

Critical distinction: df["Math"] returns a **Series**, while df[["Math"]] returns a **DataFrame**. Double brackets are also used when selecting multiple columns.

6. loc vs iloc

Method	Selection style	Example
loc	Labels / names	df.loc[1, "City"]
iloc	Integer positions	df.iloc[0, 2]

Memory shortcut: loc = label-based; iloc = integer-position-based.

7. Boolean Filtering

```
df["Math"] > 80
```

This creates a Boolean Series (True/False) called a mask.

```
df[df["Math"] > 80]
```

Passing that Boolean mask inside df[...] returns only rows where the condition is True.

8. Multiple Conditions

```
df[(df["Math"] > 80) & (df["Python"] > 90)]
```

Use & for AND and | for OR with Pandas conditions. Put each comparison inside parentheses.

9. Adding a Calculated Column

```
df["Average"] = df[["Math", "Python", "AI"]].mean(axis=1)
```

axis=1 calculates across columns for each row. Here it creates one average score per student.

10. Basic Statistical Analysis

```
df.describe()
df["Math"].mean()
df["Math"].max()
df["Math"].min()
```

These functions help summarize numerical data and identify central or extreme values.

Day 2 Practice Highlights

- Created Series with custom indexes.
- Created student DataFrames from dictionaries.
- Selected single and multiple columns.
- Used loc and iloc for row/cell selection.
- Filtered Math, Python, and AI scores using conditions.
- Combined multiple Boolean conditions.
- Created an Average column with mean(axis=1).
- Filtered students based on calculated Average values.

- Used describe(), max(), and min() for basic analysis.

Day 2 Mini-Project - Student Performance DataFrame Analyzer

- Stored student names and Math/Python/AI scores in a DataFrame.
- Calculated each student's Average.
- Filtered students using subject-score conditions.
- Generated descriptive statistics.
- Identified highest and lowest student averages.
- Practiced the Series vs DataFrame distinction.

Quick Syntax Cheat Sheet

```
pd.Series(...)
pd.DataFrame(...)
df.shape
df.columns
df.index
df.dtypes
df.head()
df.tail()
df.info()
df.describe()

df["Math"]
df[["Math"]]
df[["Name", "Age"]]

df.loc[...]
df.iloc[...]

df[df["Math"] > 80]
df[(condition1) & (condition2)]

df["Average"] = df[["Math", "Python", "AI"]].mean(axis=1)
```

Key Takeaways

- A Series is 1D; a DataFrame is 2D.
- df["column"] returns a Series; df[["column"]] returns a DataFrame.
- loc selects using labels; iloc selects using integer positions.
- A comparison such as df['Math'] > 80 creates a Boolean mask.
- Use that mask inside df[...] to filter rows.
- Use parentheses around conditions when combining them with & or |.
- axis=1 means calculate across columns for each row.
- Pandas provides the foundation for cleaning, EDA, and visualization.

Progress

Week 2 Day 2 - Pandas Fundamentals: COMPLETE
Next completed topic: Week 2 Day 3 - Data Cleaning.